SAM4s performance across the transmission path
(non-exhaustive list)

Electric Grid

Power Quality
Current PQ, harmonic distortion, volt. unbalance

VFD[1]

Motor side
Current PQ, harmonic distortion, volt. unbalance

SAM4 can detect failures in the electric grid in a direct on line set-up. With a VFD, SAM4 can detect failures between the VFD and the electric machine only.

Electric Machine

Rotor
Eccentricity, damaged rotor bar

Stator
Turn-to-turn short circuit, winding looseness

Bearings
Cage/raceway deformation

Transmission

Belt
Pulley unbalance (dirt, eccentricity)

Chain
Chain wheel unbalance (eccentricity)

Cardan shaft
Excessive clearance (e.g., screw, cross-link)

Coupling
Coupling eccentricity, unbalance

Gearbox
Broken/cracked gear teeth

Load (Asset)

- SAM4 can detect a wide variety of failures accurately and timely across different types of assets
- SAM4 has achieved excellent performance in the field with customers for key assets. Examples are pumps, fans, conveyors, rolls (or presses), and mixers. For others, its performance depends on the failure mode and application.
- Failure modes SAM4 performs (very) well on across applications:
  - Mechanical unbalance
  - Misalignment
  - Impeller (or blade) damage and condition

- Failure mode where SAM4's performance depends on the application:
  - Bearing degradation

Legend: Performance of SAM4

Situation dependent	(Very) High

SAM4 detection capabilities (1/2)

Direct cause of downtime	Rootcause of failure	Indicator of rootcause	Detectability by SAM4
1 VFD failure	VFD component degradation	Power Quality indicators (THD, Crest factor), increase in spectral energy	SAM4 calculates THD, Crest Factor and monitors increase in spectral energy
2 Motor Stator	Insulation degradation	Overcurrent	SAM4 directly measures current*
	Insulation degradation	Electrical unbalance	SAM4 calculates electrical unbalance*
	Insulation degradation	Contaminants	Not detectable by SAM4
	Insulation degradation	Environmental factors(Heat, Moisture)	Not detectable by SAM4
	High resistance connections	Voltage unbalance	SAM4 calculates voltage unbalance
3 Motor bearing failure	Mechanical unbalance/overloading	Eccentricity in the rotor, rotational frequencies	SAM4 calculates the eccentricity
	Misalignment/Incorrect installation and fit	Eccentricity in the rotor, rotational frequencies	SAM4 calculates the eccentricity
	Electrical damge	Bearing currents	Under development
	Grease degradation/misuse	Change of colour	Not detectable by SAM4
	Grease degradation /misuse	Change of smell	Not detectable by SAM4
	Overheating	Temperature, grease degradation	Not detectable by SAM4
	Improper lubrication	Eccentricity in the rotor	SAM4 calculates the eccentricity, possible if there is a excessive vibration
	Improper lubrication	Sound	Not detectable by SAM4
	Improper lubrication	Temperature	Not detectable by SAM4
4 Direct coupling failure	Mechanical unbalance/excessive motor load	Eccentricity in the rotor, rotational frequencies	SAM4 calculates the eccentricity and rotational frequencies
	Misalignment	Eccentricity in the rotor, rotational frequencies	SAM4 calculates the eccentricity and rotational frequencies
5 Gearbox failure	Mechanical unbalance	Eccentricity in the rotor, rotational frequencies	SAM4 calculates the eccentricity and rotational frequencies
	Misalignment	Eccentricity in the rotor, rotational frequencies	SAM4 calculates the eccentricity and rotational frequencies
	Geartooth degradation	Eccentricity in the rotor, rotational frequencies related to the gearbox	SAM4 calculates the eccentricity and rotational frequencies related to the gearbox
6 Belt system failure	Mechanical unbalance	Eccentricity in the rotor, rotational frequencies related to the belt	SAM4 calculates the eccentricity and rotational frequencies related to the belt
	Misalignment	Eccentricity in the rotor, rotational frequencies related to the belt	SAM4 calculates the eccentricity and rotational frequencies related to the belt

SAM4 detection capabilities (2/2)

	Direct cause of downtime	Rootcause of failure	Indicator of rootcause	Detectability by SAM4
7	Application bearing degradation	Mechanical unbalance/overloading	Eccentricity in the rotor, rotational frequencies	SAM4 calculates the eccentricity
		Misalignment/Incorrect installation and fit	Eccentricity in the rotor, rotational frequencies	SAM4 calculates the eccentricity
		Electrical damge	Bearing currents	Under development
		Grease degradation/misuse	Change of colour, smell, high temperature	Not detectable by SAM4
		Overheating	Temperature, grease degradation	Not detectable by SAM4
		Improper lubrication	Sound, high temperature	Not detectable by SAM4
		Improper lubrication	Eccentricity in the rotor	SAM4 calculates the eccentricity, possible if there is a change in vibration
8	Impeller failure	Mechanical unbalance	Eccentricity in the rotor, rotational frequencies related to the impeller	SAM4 calculates the eccentricity and rotational frequencies related to the impeller
		Misalignment	Eccentricity in the rotor, rotational frequencies related to the impeller	SAM4 calculates the eccentricity and rotational frequencies related to the impeller
		Cavitation	Spectral Energy increase around supply frequency	SAM4 calculates scorers that monitor this increase
		Clogging	Eccentricity in the rotor, rotational frequencies related to the impeller	SAM4 calculates the eccentricity and rotational frequencies related to the impeller
9	Blade failure	Mechanical unbalance	Eccentricity in the rotor, rotational frequencies related to the blade	SAM4 calculates the eccentricity and rotational frequencies related to the blade
		Misalignment	Eccentricity in the rotor, rotational frequencies related to the blade	SAM4 calculates the eccentricity and rotational frequencies related to the blade
10	Vane failure	Mechanical unbalance	Eccentricity in the rotor, rotational frequencies related to the vane	SAM4 calculates the eccentricity and rotational frequencies related to the vane
		Misalignment	Eccentricity in the rotor, rotational frequencies related to the vane	SAM4 calculates the eccentricity and rotational frequencies related to the vane
		Cavitation	Spectral Energy increase around supply frequency	SAM4 calculates scorers that monitor this increase
11	Conveyor belt failure	Misalignment	Eccentricity in the rotor, rotational frequencies related to the belt	SAM4 calculates the eccentricity and rotational frequencies to the belt
		Excessive load	Eccentricity in the rotor, rotational frequencies related to the belt, active	SAM4 calculates the eccentricity, rotational frequencies to the belt and active power
		Abrasive wear and tear	Eccentricity in the rotor, rotational frequencies related to the belt	SAM4 calculates the eccentricity and rotational frequencies to the belt

Indicative pre-warning time (non-exhaustive list)

Component	Failure mode or cause of failure	Typical pre-warning time
Supply	Voltage/current unbalance	months
	Current/voltage harmonic distortion	months
	PQ issues	months
	Voltage drops/overvoltage	months
Motors	Stator shorts (interturn/turn-to-turn)	weeks
	Stator winding looseness	weeks
	Electrical unbalance	months
	Broken/loose rotor bars	months
	Rotor eccentricities	months
	Misalignment	weeks
	Soft foot	weeks
	Bearing degradation	weeks
	Mechanical unbalance	weeks
Coupling/ gears	Coupling eccentricity	weeks
	Coupling unbalance	weeks
	Broken/cracked gear teeth	months
	Gear misalignment/eccentricities	weeks
	Pulley unbalance	months
	Belt/chain wear	months

Component	Failure mode or cause of failure	Typical pre-warning time
Pumps	Cavitation	months
	Mechanical unbalance	months
	Clogging	N/A
	Impeller damage	months
Conveyors	Misalignment	weeks
	Bearing issues rolls	*weeks*
	Belt degradation	Instantly
	Belt misalignment	N/A
Blowers/fans	Impeller damage	months
	Misalignment	weeks
	Mechanical unbalance	months